Analysis of inter-individual T-cell repertoire diversity in the three-spined stickleback during infection as a natural model system for adaptive immunity

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Introduction

- A functional T-cell repertoire of diverse T cell receptors (TCRs) is key for adaptive immunity.
- Yet, little is known regarding the degree of **natural variation** in the inter-individual diversity of the T cell repertoire, especially during **infection.**
- **AIM:** to examine the qualitative and quantitative variation and dynamics of TCRβ repertoire within and among individuals during infection.

Photo credit: Martin Kalbe, 2011

Methods

- We used the three-spined stickleback (***Gasterosteus aculeatus***) and the cestode ***Schistocephalus solidus*** as an experimental infection model system.
- 120 **lab-bred** sticklebacks from 6 families were exposed to the parasite (42 developed an infection), and 60 served as control.
- Total spleen RNA was extracted, followed by **TCRβ-targeted** 5'RACE, sequencing & data analysis.

Results

1 Repertoire clonality and publicity

- No significant differences in repertoire diversity between groups (mean of 264-320 clonotypes/repertoire)
- The majority of repertoire space is occupied by singletons, with the presence of few expanded clonotypes in all groups (Fig. 1A).
- Public clonotypes are, on average, more abundant in their repertoires than non-shared clonotypes (Fig. 1B).

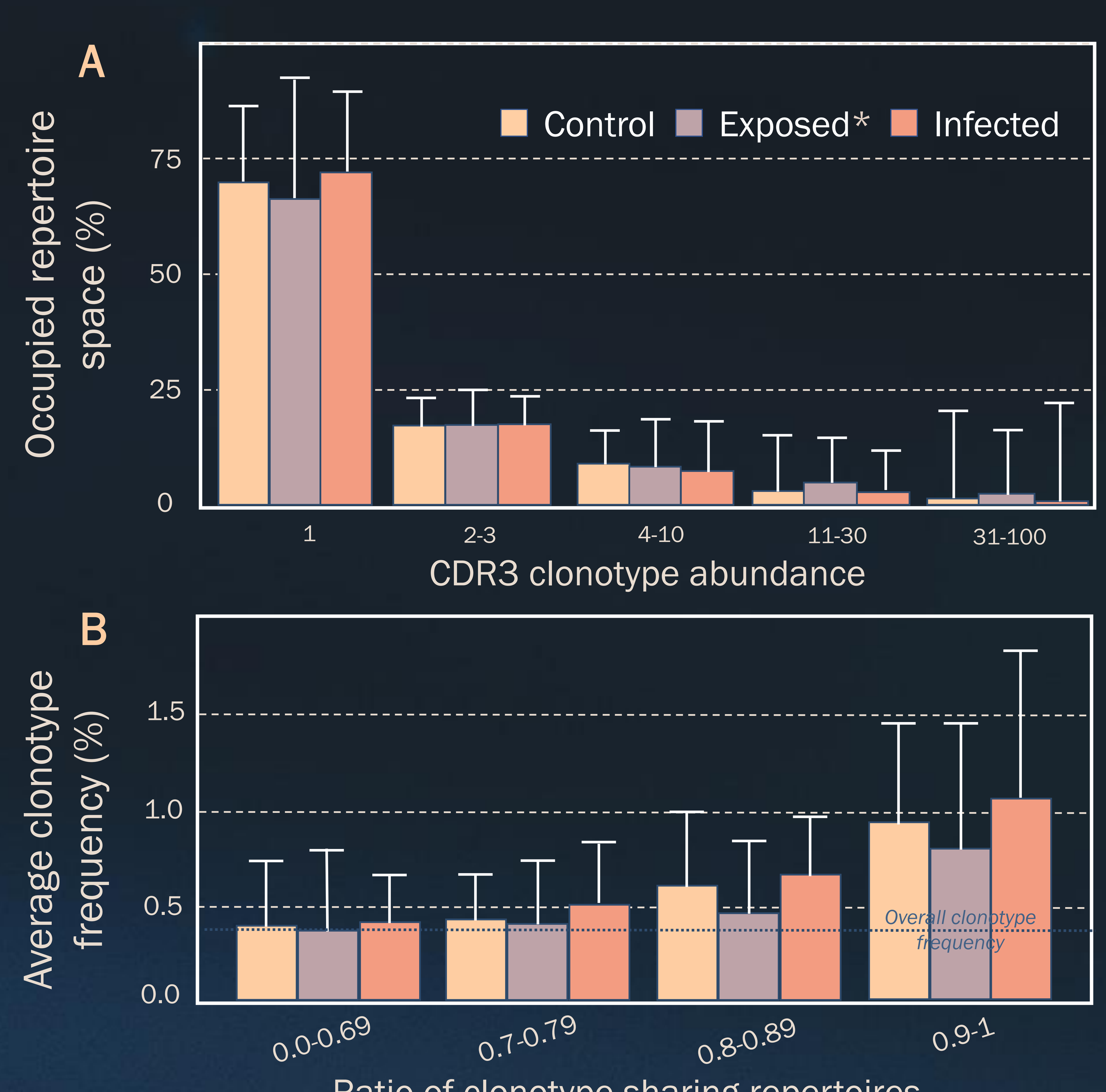


Fig. 1 | A) Distribution of mean CDR3 clonotype abundance as percentage of occupied repertoire space for each group. B) Mean per repertoire relative frequencies of clonotypes shared among increasing ratios of total repertoires in each group. Error bars indicate SD. *Exposed are those individuals who did not develop an infection after parasite exposure.

2 Repertoire similarity and clonotype overlap

- Infected individuals exhibit higher in-group repertoire similarity compared to the control group (Fig. 2A).
- Repertoires of infected individuals co-cluster together based on MDS coordinates (Fig. 2B), with post-hoc permutation tests confirming a significant effect of infection status on clustering.

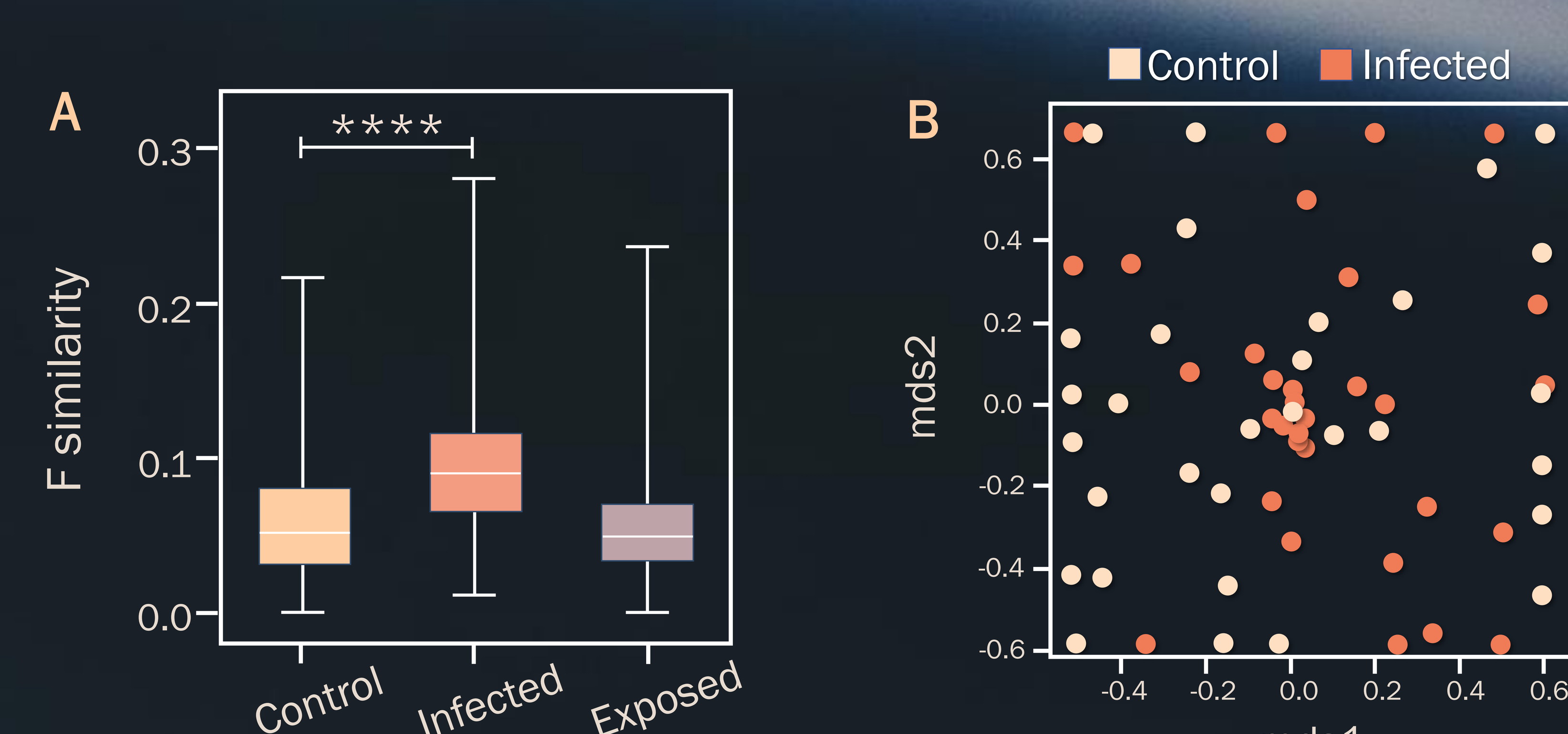
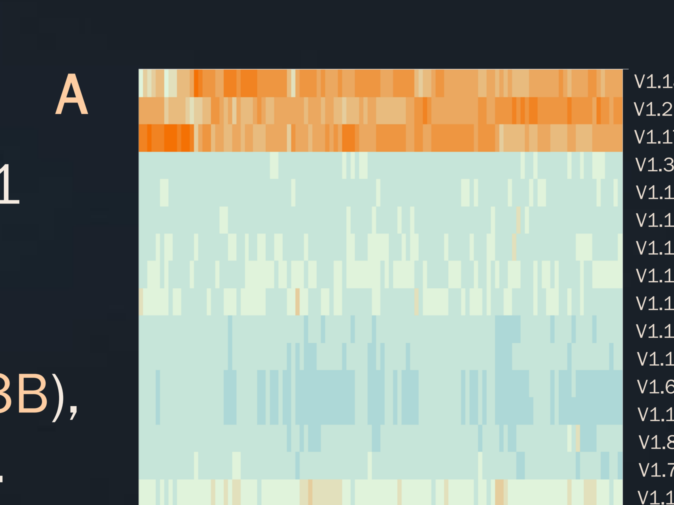


Fig. 2 | A) Box plot of F similarity metric (geometric mean of relative frequencies of all overlapping clonotypes between 2 repertoires) within each group. Horizontal line marks the mean, error bars show SD. Two-tailed P values from unpaired t-tests, ****P<0.0001. B) MDS plot of infected and control samples. Euclidean distance between points reflects the dissimilarity between repertoires.

3 V gene segment usage

- V segment usage is biased towards 3 out of the total 17 genes in the TCRβ1 repertoire (Fig. 3A).
- There is a strong family effect on the frequency of usage of V1 genes (Fig. 3B), but no such effect for infection status.



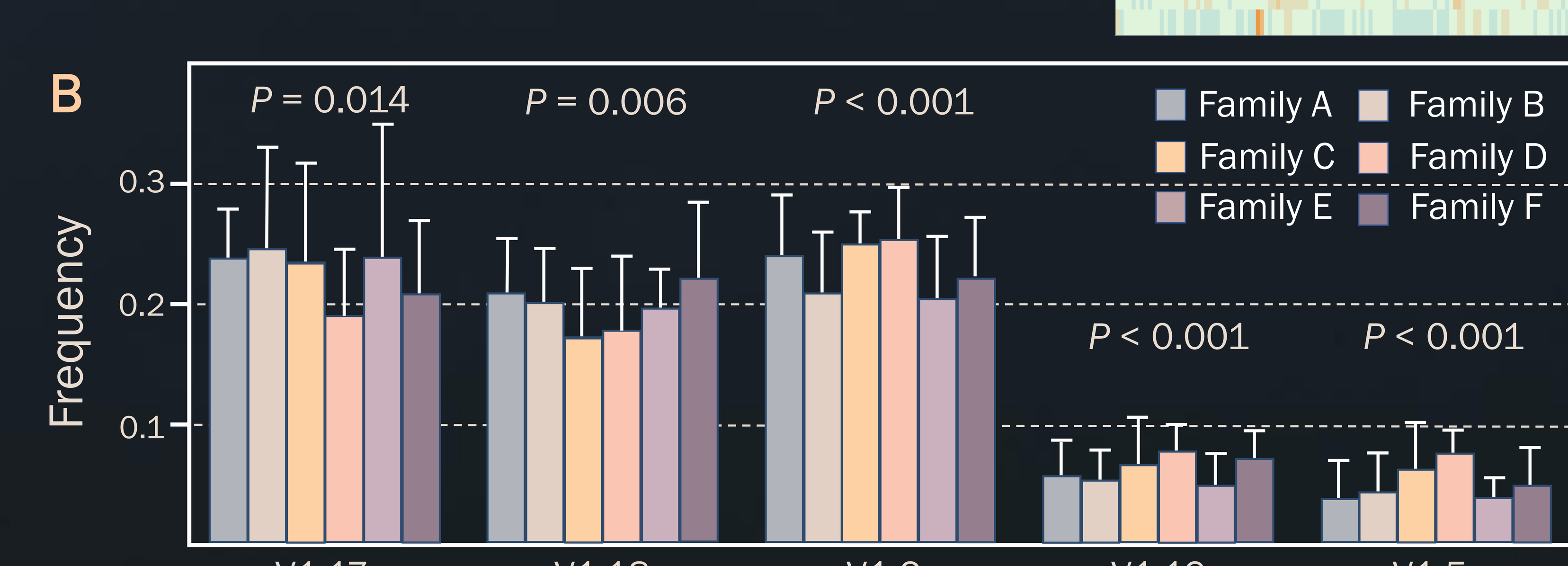


Fig. 3 | A) Heat-map of V gene segment usage frequency across all samples in the TCRβ1 repertoire. B) Frequency of usage of the top 5 V1 gene segments. P values from Kruskal-Wallis test for multiple groups, adjusted using Holm-Bonferroni correction.

Conclusions

- TCRβ repertoire profiles reveal **activation** of adaptive immunity (Fig. 1A), with expansion of **public clonotypes** hinting at non-random clonal T- cell expansion (Fig. 1B), possibly the result of **antigen-specific** responses.
- Greater sharing of prevalent public clonotypes among infected individuals explains higher repertoire **similarity** (Fig. 2).
- V segment usage analysis reveals strong **family effect** (Fig. 3) but, surprisingly, no differences based on infection status.